

Dev Assignments — Integration Phase II

Next round of improvements, features & integrations per area of responsibility, sized for free-tier limits and an early dev stage.

Date **2026-07-14** Baseline **main @ 87254e8** (PR #24 merged) Owners **A1 Session Brain** · **A2 Rec Engine** · **A3 DNA Writer**

1,853 CATALOG TITLES (12% OF 15K)	100% ENRICHED · EMBEDDED · POSTERS	90.3% TRAILER COVERAGE	12.2% CREW LINEAGE (9,704 CREW)	197 TASTE SIGNALS · 4 USERS
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Repo-wide type-check is now clean (the old @google/genai errors are resolved). Nightly grow/enrich/trailer automation is healthy — crew lineage is the one pipeline that can't catch up at its current drain rate.

■ **ASSIGNMENT 1 — SESSION BRAIN**

src/modules/session · conversation, voice, app shell

Surface the 48-hour regret prompt P0 Effort S

The component is built but never rendered. RegretPrompt.tsx and the localStorage queue (regret-queue.ts) shipped with the engine work; RecCard already enqueues on "Watched it". The product's regret dimension currently collects zero data.

On app load in **WTWApp**, read pending checks and render the prompt above the onboard view; wire glad/neutral/regret to the existing POST /feedback actions (regret / glad_watched already handled server-side). Review: A2.

src/app/components/RegretPrompt.tsx · src/lib/regret-queue.ts · src/modules/session/components/WTWApp.tsx

Link the DNA profile page from the hamburger menu P0 Effort S

The page exists; nothing links to it. /profile/dna and GET /api/dna/summary are both live, but the menu has no entry — the last unchecked item in checklist §2.

Add a **"Taste profile"** item to the hamburger (same pattern as "Your ratings"). Coordinate in the group chat — this is shared src/app/ surface. Review: A3.

src/modules/session/components/WTWApp.tsx · src/app/profile/dna

Voice sample drip + enable previews P1 Effort S

The 30 voice-picker preview buttons ship disabled until public/voice-samples/*.wav exists.

Run npm run generate-voice-samples over ~3 days, then drop the disabled attribute.

Quota-gated: Gemini free tier ≈ 10 generations/day — schedule the drip, don't burn a day's voice budget in one run (voice mode shares it).

scripts/generate-voice-samples.mjs · WTWApp.tsx (voicePlay)

Voice → Recommend handoff with query mode P1 Effort M

Exiting voice via "Recommend" re-enters the recs view generically; the user's spoken intent ("something short and funny") is dropped instead of steering the feed.

Carry the last voice turn as SessionContext.immediate_request into /api/session/end, and agree a query param on generate with A2 (their side is P1 below).

src/modules/session/voice/VoiceMode.tsx · src/app/api/session/end/route.ts

"Your ratings" follow-through

P2

Effort S

The new screen (PR #24) is functional but minimal; two cheap wins surfaced during review.

Toast/undo after re-rate & restore; empty-bucket CTA that deep-links into Fast Learning. No new API surface, zero external calls.

src/modules/session/ratings/RatingsView.tsx

■ ASSIGNMENT 2 — RECOMMENDATION ENGINE

src/modules/engine · scoring, enrichment, co-watch

Fix the co-watch IDOR (H1 — security)

P0

Effort M

Any authed user can read another user's taste profile by POSTing a guessed `user_id_b` to `/cowatch` — there is no room-membership check, and the engine loads B's DNA with the service-role client, bypassing RLS. Open since the 2026-07-01 review.

Make rooms real: a `cowatch_rooms` table (or Redis entry) where both users **join by room code**; the route then derives user B from the room record instead of trusting the request body. Reject non-members with 403.

src/app/api/recommendations/cowatch/route.ts · src/modules/engine/cowatch/intersection.ts

Unstick the crew-lineage backlog

P0

Effort M

8,500 of 9,704 crew lack lineage graphs and the gap is widening — the catalog grew ~8× while the nightly drain stayed at ~20/night (~14 months to clear, ignoring growth). Lineage-boost scoring is silently inert for most titles.

Move lineage enrichment onto the Mistral workhorse with a rate-aware batch loop (target 200–300/night), and **prioritise crew that actually appear in user signals** so scoring quality improves where it matters first.

Fits free tier: Mistral 50K TPM / 50 req-min has no daily wall — this is exactly the budget the enrichment pipeline already exploits. Avoid Groq/Gemini here.

src/modules/engine/enrichment/build-lineage-graph.ts · nightly-enrichment.ts · scripts/grow-catalog.mts

Feed the regret signal into scoring

P1

Effort M

Once A1 surfaces the prompt, regret data starts flowing — but no scoring component consumes it. Regret is designed as its own fingerprint dimension, distinct from "disliked".

Add a regret-aware penalty in visceral/narrative matching (down-weight titles similar to regretted watches), and log its contribution in `reason_payload` so "Why this?" stays honest.

src/modules/engine/scoring/* · pipeline/step6-reason-payload.ts

Explicit query mode on generate

P1

Effort M

Counterpart to A1's voice handoff: the pipeline personalises from the fingerprint but can't take a one-shot request ("90s thriller tonight") as a hard steer.

Accept `SessionContext.immediate_request` in the pipeline; embed the request via Mistral and blend it into candidate scoring for this generation only (`session_override_active`), without touching stored DNA.

1 embed call per steered generation — bounded, cache by request text.

src/modules/engine/pipeline/generate.ts · step2/step3 scoring

Stretch-pick outcome audit

P2

Effort S

Stretch picks (1-in-20, suppressed <15 signals) now have enough live feedback to sanity-check; accept/reject is itself a signal but nobody has looked at the data.

Small admin script reading `stretch_pick_history` across users: acceptance rate, reaction split, whether suppression triggers correctly. Output informs tuning, no product change yet.

src/modules/dna (read-only) · scripts/

writeDNA end-to-end verification P0 Effort M

Checklist §3 is still open: the full write pipeline has never been proven against live Supabase with a **real** SessionSummary (only unit-level). With 4 real users accumulating signals, silent merge bugs now corrupt real fingerprints.

Scripted E2E on a dev user: bootstrap → synthetic session with known reactions → assert strand A/B/C deltas, taste_version bump, snapshot write, embedding upsert, Redis invalidation. Keep it re-runnable as the regression harness.

src/modules/dna/writer.ts · scripts/ (new) · supabase dev instance

Close the embedding-ref contract with the engine P1 Effort S

Checklist §1's last open seam: the fingerprint_embeddings upsert format was never formally confirmed between writer and engine — it works today by inspection, not by contract.

Write the shape down as a type in src/types/, add one integration assert to the E2E harness above. Review: A2.

src/modules/dna/embedding.ts · src/modules/engine/scoring/narrative-match

Re-tune reaction weights for the 3-level enum P1 Effort S

PR #24 retired "mixed"; ambivalent chat reactions now map to disliked (−0.20 crew delta vs mixed's old −0.05).

Lukewarm ≠ negative — the fingerprint may now over-penalise mild reactions from transcripts.

Decide: either soften disliked deltas for transcript-sourced signals (confidence-weighted), or have the session analyzer drop lukewarm mentions entirely. Small change, real fingerprint impact — needs a deliberate call, not a default.

src/modules/dna/lib/reaction-score.ts · update-strand-c.ts · src/modules/session/analyze-session.ts (A1 review)

Temporal-decay dry run P1 Effort S

The 4am decay cron has been live for weeks but no signal is older than ~2 months — the 18-month/50% path and the 30-day run guard have never executed on real-shaped data.

Seed a dev user with back-dated signals, run applyTemporalDecay directly, assert weights and the guard. Cheap now; painful to debug in 16 months.

src/modules/dna/temporal-decay.ts · /api/cron/decay

Fingerprint history on the profile page P2 Effort M

Snapshots (last 5, with rollback) exist but are invisible; the DNA summary page A1 is linking (their P0) is the natural home.

Show taste_version history with per-snapshot diffs ("Nolan affinity ↑, slow-burn ↓") and a guarded rollback. Pure Supabase reads — no LLM cost.

src/modules/dna/snapshot.ts · src/app/profile/dna (coordinate with A1)

Full-flow E2E smoke (checklist §3) P0 Effort M · A1 + A3 pair

One scripted pass with real keys: bootstrap DNA → chat → session end → generate → assert Redis cache hit by taste_version → rate → regret. Automate as far as OAuth allows (service-role test user below the UI). This is the gate before inviting more test users.

Test-user seeding kit

P2

Effort S · any

4 users is too thin to evaluate scoring. A small script that seeds N synthetic users with distinct taste archetypes (auteur-lover, comfort-watcher, genre-hopper) gives A2/A3 evaluation data without waiting on real sign-ups.

FREE-TIER GUARDRAILS

standing rules for every assignment above

PROVIDER	USE FOR	LIMIT THAT BITES	RULE
Mistral (embed key)	Enrichment, structured output, embeddings — the workhorse	50K TPM · 50 req/min, no daily wall	Route all new batch/structured jobs here (lineage, query-mode embeds).
Groq gpt-oss-120b	Chat & greetings only	Reasoning model — breaks generateObject (empty content)	Never use for structured output; model IDs only via <code>src/lib/ai-models.ts</code> .
Gemini Live	Voice mode + voice samples	≈ 10–20 req/day	Nothing new on Gemini; sample generation is a multi-day drip.
TMDB / OMDb	Metadata at enrichment time	Generous, but per-request	Read from cached <code>titles</code> columns at serve time — never re-fetch per request (poster CDN images are keyless and free).
Upstash Redis	Rec cache, DNA cache	Free-tier command quota	Cache keyed by <code>taste_version</code> ; no per-card LLM regeneration (decision from the ratings-screen review).

SUGGESTED SEQUENCING

SPRINT 1 — CLOSE THE LOOP

A1

Regret prompt surfaced

A1

Profile link in menu

A2

Co-watch IDOR fix

A3

writeDNA E2E harness

SPRINT 2 — QUALITY & STEER

A2

Lineage backlog drain

A2

Query mode on generate

A1

Voice handoff + samples

A3

Reaction re-tune · decay dry run

SPRINT 3 — DEPTH

A2

Regret-aware scoring

A3

Fingerprint history UI

AII

Full-flow E2E + seeding kit

A2

Stretch-pick audit